GLOBAL OLYMPIC SPORTS PERFORMANCE ANALYSIS

PROJECT DOCUMENTATION

Data Visualization Using Tableau Public

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ABSTRACT:

The Olympic Games represent one of the world's largest international sporting events, bringing together athletes from different nations and sporting disciplines. The analysis of Olympic data can provide valuable insights into medal distribution, participation patterns, and the representation of countries and sports across recorded Olympic events.

The project Global Olympic Sports Performance Analysis uses Tableau Public to transform Olympic athlete-event data into meaningful visualizations. The dataset, named "athlete_events.csv", is explored through three major visualizations: Olympic Medal Distribution, Olympic Participation by Country, and Sports Participation.

The visualizations are combined into an interactive dashboard to enable convenient comparison of different categories. A Tableau Story is also included to communicate the observations in a structured and understandable sequence.

The project demonstrates the practical application of data visualization in exploring structured datasets, identifying descriptive patterns, and presenting information in a visually accessible format.

1. INTRODUCTION:

Data visualization is an important technique for converting complex datasets into meaningful visual representations. Large datasets may contain thousands of records, making it difficult to identify patterns through manual inspection alone. Charts and dashboards simplify this process by presenting information in a clear and organized manner.

The Olympic Games provide an interesting subject for data analysis because they involve multiple countries, sports, athletes, events, and medal categories. Examining these records

can help viewers understand how participation and medal-related records are distributed across different categories.

This project focuses on analyzing Olympic athlete-event data using Tableau Public. Three worksheets are developed and combined into a dashboard, supported by a Tableau Story. The resulting visualization provides an accessible overview of selected patterns in the dataset.

2. PROBLEM STATEMENT

Olympic datasets contain numerous records and categories that can be difficult to interpret when viewed in their original tabular format. Comparing medal categories, identifying sports with high record counts, and examining the representation of different National Olympic Committees can require considerable effort when performed manually.

The problem addressed by this project is how to transform Olympic data into clear, meaningful, and easy-to-understand visualizations that support efficient comparison and exploration.

The project addresses this challenge by using Tableau Public to organize the data into charts, a consolidated dashboard, and a Story that presents the main observations systematically.

3. OBJECTIVES OF THE PROJECT:

The primary objectives of this project are:

1. To analyze the distribution of recorded Olympic medals across medal categories.
2. To compare record counts across different sports.
3. To identify the National Olympic Committees with the highest displayed record counts.
4. To create clear and informative visualizations using Tableau Public.
5. To combine multiple worksheets into a single dashboard.
6. To present the analysis through a Tableau Story.
7. To develop practical skills in data visualization, dashboard design, and analytical interpretation.
8. To communicate the results in a structured and professional manner.

4.DATASET DESCRIPTION:

The project uses a dataset named "athlete_events.csv", which is connected to Tableau Public as the source of the visualizations.

The commonly distributed Olympic athlete-events dataset contains information relating to athletes, Olympic Games, sports, events, teams, National Olympic Committee codes, and medal categories. The exact fields and historical coverage depend on the version of the dataset used.

Typical fields in this dataset may include:

- Name: Athlete's name.
- Sex: Recorded sex category.
- Age: Athlete's age, where available.
- Height and Weight: Physical attributes, where available.
- Team: Team name associated with the record.
- NOC: National Olympic Committee code.
- Games: Olympic Games edition.
- Year: Year of the Olympic Games.
- Season: Summer or Winter.
- City: Host city.
- Sport: Sporting discipline.
- Event: Specific event within a sport.
- Medal: Medal category, such as Gold, Silver, or Bronze.

The fields actually available in the project file should be verified before the report is submitted.

Data interpretation note: A record in an athlete-event dataset may represent an athlete's participation in an event rather than a unique individual. Consequently, the number of records should not automatically be interpreted as the number of unique athletes or unique medals.

5. TOOLS AND TECHNOLOGIES USED:

5.1 Tableau Public

Tableau Public is the primary visualization tool used in this project. It enables the creation of charts, dashboards, and Stories through a graphical interface. It is also used to publish the visualization online.

5.2 CSV Dataset

The file "athlete_events.csv" serves as the source of the Olympic data used in the analysis.

5.3 Tableau Worksheets

Worksheets are used to create individual visualizations for medal distribution, country-level representation, and sports participation.

5.4 Tableau Dashboard

The dashboard combines the individual worksheets into one visual interface, allowing users to examine multiple aspects of the data together.

5.5 Tableau Story

The Story provides a structured presentation of the visualizations and helps communicate the key observations.

6. METHODOLOGY:

The project follows a systematic data visualization workflow.

Step 1: Data Import

The "athlete_events.csv" dataset is connected to Tableau Public so that its fields can be used to create visualizations.

Step 2: Data Exploration

Relevant fields, including Medal, NOC, and Sport, are identified for the analysis. The available measures and dimensions are reviewed before creating the charts.

Step 3: Worksheet Creation

Three worksheets are developed:

- Medal Analysis
- Olympic Participation by Country
- Sports Participation

Each worksheet focuses on a different aspect of the dataset.

Step 4: Data Visualization

Bar charts are used to compare categories and make differences in record counts easier to interpret. Category labels and numeric values are displayed to support comparison.

Step 5: Dashboard Development

The three worksheets are combined into a dashboard titled Global Olympic Sports Performance Analysis. The layout places Sports Participation in the upper-left area, Olympic Participation by Country in the lower-left area, and Olympic Medal Distribution on the right.

Step 6: Story Development

A Tableau Story is used to present the analysis in a structured sequence and explain the observations from the visualizations.

Step 7: Publication and Verification

The workbook is published to Tableau Public. The published dashboard and Story should be checked to ensure that the charts display correctly and the online link is accessible.

7. VISUALIZATION 1: OLYMPIC MEDAL DISTRIBUTION:

7.1 Purpose

The Olympic Medal Distribution visualization compares the displayed record counts for Gold, Silver, and Bronze medals. It provides a quick overview of how the three medal categories are represented in the current view.

7.2 Values Observed

Medal Category| Displayed Count
Silver| 13,372
Gold| 13,295
Bronze| 13,116

Note: These values are transcribed from the supplied Tableau dashboard screenshot and should be checked against the final workbook.

7.3 Interpretation

Silver has the highest displayed count at 13,372, followed by Gold at 13,295 and Bronze at 13,116. The differences between the three displayed counts are relatively small.

The visualization makes these differences easier to identify than examining the source data manually. It also provides a compact summary of the medal categories included in the current view.

The displayed figures should be interpreted according to the aggregation and filters applied in Tableau. Unless the workbook explicitly calculates unique medals, these values should be described as record counts by medal category rather than as definitive totals of unique medals.

8. VISUALIZATION 2: OLYMPIC PARTICIPATION BY COUNTRY:

8.1 Purpose

This visualization compares the displayed record counts associated with the National Olympic Committee (NOC) codes included in the chart. It highlights the ten NOCs shown in the current dashboard view.

8.2 Values Observed

NOC Code| Displayed Count
USA| 18,853
FRA| 12,758
GBR| 12,256
ITA| 10,715
GER| 9,830
CAN| 9,733
JPN| 8,444
SWE| 8,339
AUS| 7,638
HUN| 6,607

Note: These values are transcribed from the supplied dashboard screenshot.

8.3 Interpretation

USA has the highest displayed record count among the ten NOCs shown, with 18,853 records. FRA and GBR follow with 12,758 and 12,256 records, respectively.

The visualization makes it easy to compare the NOCs included in the chart and observe differences in their representation within the selected dataset.

However, these counts do not independently establish the population size of a country, the athletic ability of its competitors, or its overall Olympic success. They describe record counts within the current data view. NOC codes may also reflect historical Olympic committee designations rather than a direct match with present-day national borders.

9. VISUALIZATION 3: SPORTS PARTICIPATION:

9.1 Purpose

The Sports Participation visualization compares the displayed record counts across sports. The bars are arranged in descending order, making it easier to identify the sports with the highest counts in the current view.

9.2 Values Observed

Sport| Displayed Count
Athletics| 38,624
Gymnastics| 26,707
Swimming| 23,195
Shooting| 11,448
Cycling| 10,859
Fencing| 10,735
Rowing| 10,595
Cross Country Skiing| 9,133

Alpine Skiing| 8,829
Wrestling| 7,154

Note: These values are transcribed from the supplied dashboard screenshot.

9.3 Interpretation

Athletics has the highest displayed record count at 38,624, followed by Gymnastics at 26,707 and Swimming at 23,195.

The difference between these three sports and the remaining sports in the displayed top ten is noticeable. The descending bar chart makes the ranking and relative differences easier to interpret.

This visualization provides a useful overview of the sports represented in the selected view. However, record counts should not be treated as unique athlete totals unless a distinct-athlete calculation has been applied.

10. DASHBOARD DESIGN AND FUNCTIONALITY:

The dashboard combines the three worksheets into one interface under the title Global Olympic Sports Performance Analysis.

The arrangement supports a comparison of three related perspectives:

- Medal-category distribution.
- Record counts by NOC.
- Record counts by sport.

Horizontal bar charts are suitable for the category-based comparisons because they allow labels to remain readable while displaying relative values. Numeric labels further assist viewers in comparing the categories.

The combined dashboard reduces the need to navigate between separate worksheets and provides a convenient overview of the selected measures.

Before final submission, the dashboard should be checked for readable titles, visible category labels, consistent formatting, and any chart elements that may be clipped on the published page.

11. TABLEAU STORY AND PROJECT DEMONSTRATION:

The Tableau Story complements the dashboard by presenting the analysis in a logical sequence. It should introduce the overall project and then guide viewers through the three visualizations.

A suitable sequence is:

1. Introduction to the Olympic dataset and the purpose of the project.
2. Explanation of the Olympic Medal Distribution visualization.
3. Comparison of the NOC record counts.
4. Discussion of the Sports Participation visualization.
5. Summary of the main observations and limitations.

The demonstration video provides a concise walkthrough of the dashboard and Story. It should show the final visualizations and explain what each chart represents.

Since a demonstration video has already been recorded, it can be used for submission if it accurately reflects the final version of the published project.

12. KEY FINDINGS:

Based on the values visible in the supplied dashboard screenshot, the following observations can be made:

Finding 1: Medal Distribution

Silver has the highest displayed medal-category record count at 13,372, followed by Gold at 13,295 and Bronze at 13,116.

Finding 2: NOC Representation

USA has the highest displayed NOC record count among the ten categories shown, with 18,853 records. FRA and GBR are the next two categories in the displayed ranking.

Finding 3: Sports Representation

Athletics has the highest displayed sports record count at 38,624, followed by Gymnastics and Swimming.

Finding 4: Value of Combined Visualization

The dashboard provides a consolidated view of three different aspects of Olympic data. The combination of worksheets makes it easier to compare categories and identify patterns.

These findings are descriptive and apply to the displayed data view. They may change if the dataset, filters, or aggregation methods are modified.

13. LIMITATIONS OF THE PROJECT:

Although the dashboard provides a useful overview, several limitations should be considered.

1. Record Count Interpretation: Counts may represent athlete-event records rather than unique athletes, events, or medals.
2. Dataset Coverage: The exact version and historical coverage of the CSV should be confirmed from the source file.
3. Aggregation and Filters: Displayed results depend on the fields, calculations, filters, and aggregation methods used in Tableau.
4. Historical NOC Codes: NOC codes may represent historical Olympic committee designations that do not directly correspond to current national borders.
5. Descriptive Nature: The visualizations reveal patterns but do not explain the causes behind differences in participation or medal records.
6. Data Quality: Missing values, duplicate records, or inconsistent categories may affect the analysis if not examined.

Recognizing these limitations helps ensure that the visualizations are interpreted accurately and responsibly.

14. FUTURE SCOPE:

The project can be improved through several additional analytical features.

- Introduce filters for year, season, sport, gender, and medal category.
- Compare Summer and Winter Olympic Games separately.
- Add year-wise trends to examine changes in participation over time.
- Use distinct athlete counts when the goal is to measure unique individuals.
- Examine data quality, including missing values and duplicate records.
- Add descriptive tooltips explaining the meaning of each measure.
- Extend the dashboard with additional comparisons, provided that the underlying data supports them.

These improvements would increase the analytical depth of the project and make the dashboard more useful for exploring specific questions about Olympic data.

15. CONCLUSION:

The Global Olympic Sports Performance Analysis project demonstrates the use of Tableau Public to transform Olympic athlete-event data into a clear and organized visual report.

The project includes three worksheets covering medal-category records, NOC representation, and sports participation. These worksheets are combined into a dashboard, while the Tableau Story and demonstration video support the communication of the analysis.

The visualizations make it easier to compare categories and identify patterns in the selected dataset. The displayed results highlight the leading categories in the current views while also demonstrating the importance of interpreting record counts according to the underlying data and aggregation methods.

Overall, the project provides practical experience in data visualization, dashboard development, and analytical presentation. Further improvements, such as time-based comparisons and more precise measures, could expand its usefulness and analytical accuracy.

REFERENCES:

1. Tableau Public — https://public.tableau.com/
2. Tableau Official Website — https://www.tableau.com/
3. Olympic athlete-events dataset — "athlete_events.csv" used as the source file for this project. Record the original dataset provider and URL here if required by your institution.

APPENDIX A: PROJECT SUBMISSION CHECKLIST

- [ ] Tableau workbook saved locally, preferably as a packaged workbook (".twbx") if available.
- [ ] Published Tableau Public link checked and accessible.
- [ ] Dashboard displays the intended charts correctly.
- [ ] Tableau Story is available and readable.
- [ ] Project documentation saved as required by the institution.
- [ ] Existing demonstration video shows the final dashboard and Story.
- [ ] Source dataset retained if required.
- [ ] All reported values checked against the final workbook.

APPENDIX B: PROJECT DETAILS

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Tableau Public Project
URL:https://public.tableau.com/app/profile/mahi.agrawal5372/viz/Global_Olympic_Sports_Performance_Analysis_17931952053120/OlympicPerformanceDashboard?publish=yes